welcome

Thank you for purchasing the MEAP version of *Bayesian Optimization in Action*!

While an incredibly interesting topic (albeit from my biased perspective), Bayesian optimization may appear elusive to many machine learning practitioners. The main culprit is the accessibility and quality of available resources: a full, in-depth treatment of the topic can only be found in textbooks and research papers, while online tutorials and blog posts lack the depth, structure, and consistency necessary for a solid understanding. I remember when I first learned about Bayesian optimization, it took me a long time to synthesize what I had read from these various sources and understand what they meant as a whole.

This book sets out to address this problem for newcomers who'd like to learn about Bayesian optimization—after all, every book is what the author wishes had existed when they first learned a topic. In the text, you will find a practical guide to what Bayesian optimization is, how it facilitates decision-making under uncertainty to optimize expensive processes, what different variations of Bayesian optimization there are, and last but not least, how to implement them with code. To be able to hit the ground running immediately, you will need a firm grasp on core concepts in machine learning, statistics, and Bayesian probability, as well as experience in Python programming.

Presented in accessible language, each chapter tells the story of each component of the Bayesian optimization framework. These components range from building a machine learning model, equipping the model with assumptions and prior knowledge, to using that model to make effective decisions. The book as a whole makes up a comprehensive narrative that starts you off from the ground up and introduces you to state-of-the-art techniques. Aiming to be a practical and hands-on resource, the text includes many code examples and exercises that will help solidify theoretical discussions with concrete use cases.

As you will see, one of the recurring themes of this book is that effective decision-making relies on intelligently collecting information relevant to our goal. The goal of this platform is to make the book better, and I rely on you, the readers, to provide the relevant information, that is, your feedback, to steer the book in the right direction. I look forward to seeing your comments, questions, and suggestions in the liveBook Discussion forum!

— Quan Nguyen

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